

# 4

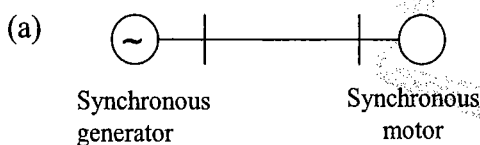
## Power System Stability

The word stability does exist in power system network due to availability of synchronous machines in the system. Most of power Generation can be made available on economical basis with the help of conventional sources by having synchronous Generator, which are located at remote places with respect to load centre. The bulk amount of power generation is able to transmit to the load center by using EHV lines.

Stability is the ability to maintain synchronism with the externally connected transmission lines by the synchronous generator in order to deliver the maximum power to either small (or) sudden change in load. The two electrical parameters for the stability consideration are

- (i) supply frequency (ii) Terminal voltage.

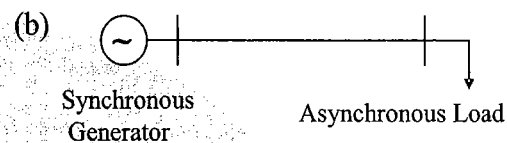
In general the real power is only considered for stability consideration, because any change in load at load end i.e., real power does reflect in change of rotor position of an alternator. Most common words in stability are acceleration / deceleration, moment of inertia, angular velocity, power angle etc. The two possibilities to fall out of synchronism (or) losing synchronism are



In the above diagram synchronous generator is connected synchronous motor through a long loss-less transmission line. Stalling of synchronous motor at its final position due to sudden and large change load on the shaft of motor.

The sudden increase in mechanical output on the motor ( $P_m$ ) may not result in increase input i.e.;  $p_e$ , because the input to the motor is same as output of the generator. The output of the generator does only increase by increasing the mechanical input of the generator. So the synchronous motor will experience deceleration, may result in fallout of

synchronous. The falling of synchronous motor will result in fall out of synchronous generator because the output of generator is  $p_e = 0$ , where as the mechanical input  $p_s$  is same. The synchronous generator will experience acceleration. So the synchronous generator connected to synchronous motor, the both machines will fall out of synchronism or both will maintain synchronism.

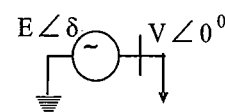


In the above diagram synchronous generator is connected to asynchronous load through a loss less long transmission line. A wild and sudden fluctuations in the voltages of the interconnected transmission line due to short-circuit fault may result in fall out off synchronism of a synchronous generator.

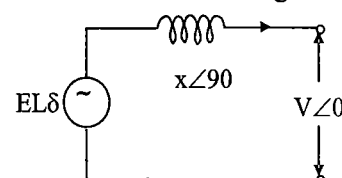
So the chance of fall out of synchronism is due to sudden and large fluctuations, but not due to small gradual variations.

**Steady State Stability:** It is the ability to maintain synchronism by delivering maximum amount of real power for a small and gradual variation of load. The rate of change of load is less than rate of change of excitation controller or the frequency of oscillations due to change of load are less than natural frequency of the system. The steady state stability limit can be evaluated by using power-transfer equations by using simulation networks.

**Ideal Case I:**



Alternator connected to asynchronous load directly. The resistance of alternator is ignored.





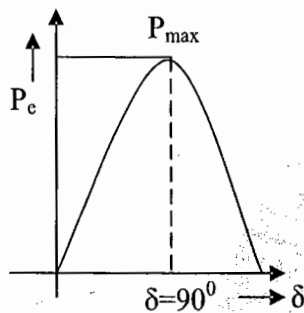
$$S = V I^*$$

$$= |V| \angle 0 \left( \frac{E \angle \delta - V \angle 0}{X \angle 90} \right)$$

$$S = \frac{EV}{X} \angle (90 - \delta) - \frac{V^2}{X} \angle 90$$

$$P_e = \frac{EV}{X} \sin \delta = \text{Real power output.}$$

$$P_e = P_{\max} = \frac{EV}{X} [\delta = 90^\circ]$$



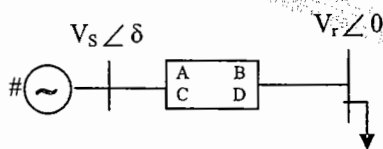
Synchronizing power will indicate the degree of magnetic coupling.

$$\frac{\partial P_e}{\partial \delta} = \frac{EV}{X} \cos \delta = P_m \cos \delta$$

$\delta > 90^\circ$  ;  $P_e$  is reducing

$$\frac{\partial P_e}{\partial \delta} \text{ is negative}$$

**Practical case:**



$$S_r = V_r I_r^* = V_r \angle 0 \left[ \frac{V_s \angle \delta - |A| \angle \alpha V_r \perp \delta}{|B| \angle \beta} \right]$$

$$P_r = \frac{|V_s| |V_r|}{|B|} \cos(\beta - \lambda) - \frac{|A|}{|B|} |V_r|^2 \cos(\beta - \alpha)$$

**Special case:** short transmission line

$$|A| = 1 \quad \alpha = 0$$

$$P = \frac{|V_s| |V_r|}{|B|} \cos(\beta - \delta) - \frac{|V_r|^2}{|B|} \cos \beta$$

$$\text{At } \delta = \beta = \tan^{-1} \left( \frac{X}{R} \right)$$

$$P = P_{\max} = \frac{|V_s| |V_r|}{|B|} - \frac{|V_r|^2}{|B|} \cos \beta$$

$$P_{\max} = \frac{(V_s)(V_r)}{\sqrt{R^2 + X^2}} - \frac{(V_r^2)}{\sqrt{R^2 + X^2}} \frac{R}{\sqrt{R^2 + X^2}}$$

$$\frac{d(P_{\max})}{dx} = 0$$

$$X = \sqrt{3}R \text{ (OR) } \frac{X}{R} = \sqrt{3}$$

**Uncompensated Transmission Line:** In short line the total value of  $x$  (reactance) is less so the alternator will deliver maximum power without losing synchronism.

**Compensated Transmission Line:** In long transmission line the total value of reactance is very high so that alternator may fall out of synchronism, while delivering maximum power. To reduce the net reactance, series capacitor is placed. Such a transmission line is called compensated line.

**Methods of improving steady state stability limit:**

$$P_{SSSL} = \frac{EV}{X}$$

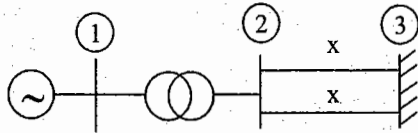
- (i) Operating the system at higher voltages
- (ii) Reducing the net reactance of the system.

This can be obtained by using

- parallel lines
- double circuit
- bundle conductors
- series capacitors
- mid point compensation

**Example 4.1 :**

A generator with constant 1.0 p.u. terminal voltage supplies power through a step-up transformer of 0.12 p.u. reactance and a double circuit line to an infinite bus bar as shown in the figure. The infinite bus voltage is maintained at 1.0 p.u. Neglecting the resistances and susceptances of the system, the steady state stability power limit of the system is 6.25 p.u. If one of the double-circuit is tripped, then resulting steady state stability power limit in p.u. will be



**Sol:** Before fault

Steady state stability power

$$\text{Limit} = \frac{EV}{X_{eq}} = 6.25$$

$$\frac{1.0 \times 1.0}{\left(0.12 + \frac{X}{2}\right)} = 6.25 \Rightarrow X = 0.08 \text{ P.u.}$$

After fault steady stability power limit

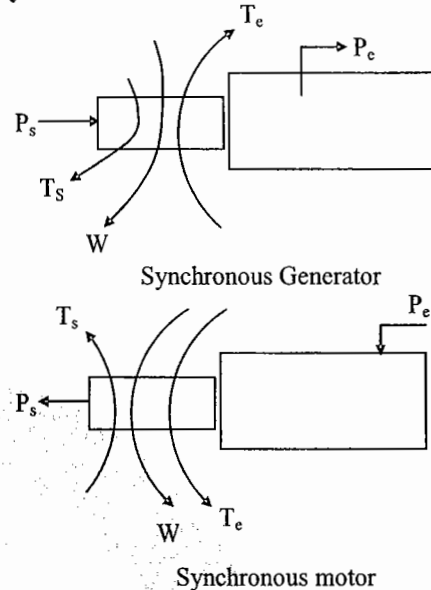
$$= \frac{1.0 \times 1.0}{0.12 + 0.08} = 5.0 \text{ P.u.}$$

**Transient Stability:**

It is the maximum amount of power that can be delivered to the load without losing synchronism for sudden and large variations of the load due to 3-phase short circuit fault occurred for a time of 5 cycles only. Otherwise the system will lose synchronism.

For the given network, the power transfer before fault is higher than that of power transfer fault. Some times the power delivery becomes zero for a fault at the generator bus terminal (or) fault on only one of the transmission line near bus bar

**1. Mechanical and Electrical powers of Synchronous Generator:**



If the Input is more than output

$$P_a = P_s - P_e \text{ (acceleration power)}$$

$$P_a = T_a \times W \text{ (W = angular velocity)}$$

$$W = \frac{2\pi N}{60}$$

N = actual speed r.p.m

$$P_a = I \propto W \text{ [ I = Inertia kg/m}^2 \text{ ]}$$

$\alpha$  = angular acceleration

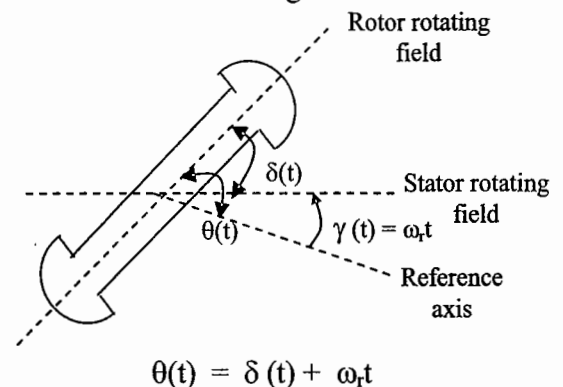
$$P_a = M \alpha \text{ [ M = Moment of Inertia (or) angular momentum]}$$

$$M = \text{J-sec / Mech. rad}$$

$$M = MJ - \text{sec / Mech. rad} \times p/2$$

$$M = MJ - \text{sec / Mech. rad} \times p/2 \times 57.11$$

$$= MJ - \text{sec / elec. degrees.}$$





[ $w_r$  = Relative angular velocity between stator and reference axis]

$$\frac{d\theta(t)}{dt} = \frac{d\delta(t)}{dt} + W_r \Rightarrow \frac{d^2\theta(t)}{dt^2} = \frac{d^2\delta(t)}{dt^2} = \alpha$$

$$\frac{Md^2\delta(t)}{dt^2} = P_a = P_s - P_e$$

$$\frac{Md^2\delta(t)}{dt^2} = P_s - \frac{EV}{X} \sin \delta$$

Swing equation which is a non-linear differential equation. It describes the relative position of rotor with respect to stator fixed as a function of time.

#### Inertia Constant:

$$(H) = \frac{\text{Kinetic energy stored}}{\text{Rating of the machine}}$$

$$(H) = \frac{(1/2)Iw^2}{S}, [M = Iw]$$

$$M = \frac{2SH}{2\pi f} = \frac{SH}{\pi f} \text{ (radians)}$$

$$M = \frac{SH}{180f} \text{ (degrees)}$$

$$M \propto H$$

The transient stability can be analyzed by using

The swing equation.

$$M \frac{d^2\delta}{dt^2} = P_s - \frac{EV}{X} \sin \delta$$

The solution for the above swing equation can be obtained as follows:

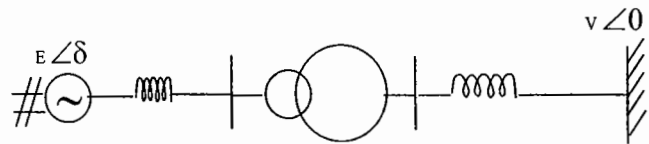
#### Swing Equation

$$M \frac{d^2\delta}{dt^2} = P_a$$

It is used to decide the stability of a synchronous machine in a Transient condition. Solution of Swing equation is required. It depends on number of syn machines.

#### Single Machine System:

- (i) Synchronous Generation is connected to an infinite bus by a loss less network.

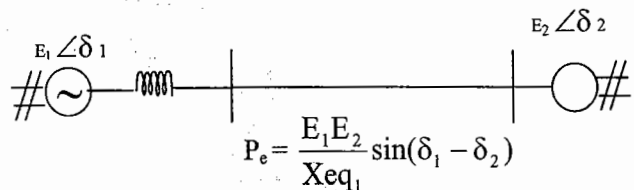


$$P_e = \frac{EV}{X_{eq1}} \sin \delta$$

Equal Area Criteria is used to get the solution of swing Equation. It is a Graphical study

#### (ii) Two machine System:

Synchronous Generation is connected to syn motor by a loss less network



$$P_e = \frac{E_1 E_2}{X_{eq1}} \sin(\delta_1 - \delta_2)$$

In a two machine system, synchronous Generator is connected to syn motor by a loss-less network. Initially both the machines in a stable condition. When the load on the motor increases suddenly, the electrical input cannot change simultaneously so that the motor will get oscillate. When the oscillations are very high when compared to natural frequency of oscillation the motor has become unstable.

When the motor has become unstable, the load on the synchronous generator is zero and the generator will make oscillations. These oscillations are also more than natural frequency of oscillations so that the generator also becomes unstable.

The above two machine system has become single machine system because the two machines are stable (or) unstable at any point of time. The equal Area Criteria is applicable.

#### (iii) Multi Machine System:

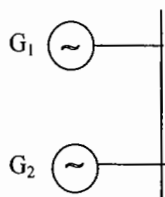
More than two machines are involved. The solution of swing equation can be obtained by point by point method (or) step by step method. It is a mathematical solution. Euler solution (or) RKU method is used.

**Note:** Though the equivalent area is stated as single-machine system it is applicable for two machine system



### Two machines are swinging together

Two alternators are connected to common bus, any change of load will change the rotor position of two alternators is called swinging of two machines.



$M_1$  and  $M_2$  are angular momentum of two generators.  $\delta_1 = \delta_2 = \delta$  and  $P_{aeq1} = P_{a1} + P_{a2}$

$$M_{eq} \frac{d^2\delta}{dt^2} = M_1 \frac{d^2\delta_1}{dt^2} + M_2 \frac{d^2\delta_2}{dt^2}$$

$$M_{eq} \frac{d^2\delta}{dt^2} = (M_1 + M_2) \frac{d^2\delta}{dt^2}$$

$$M_{eq} = M_1 + M_2$$

$$M_{eq} = M_1 + M_2 + \dots + M_n$$

### Two machines are not swinging together

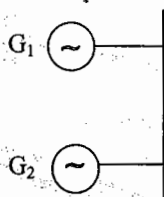
When two alternators are not connected to common bus, only change of load will change the rotor position of one alternator only, then the machines are not swinging together.

$M_1$  and  $M_2$  are angular momentum of two generators.

$$\delta = \delta_2 - \delta_1$$

$$\frac{d^2\delta}{dt^2} = \frac{d^2\delta_2}{dt^2} - \frac{d^2\delta_1}{dt^2}$$

$$\frac{d^2\delta}{dt^2} = \frac{P_{s2} - P_{e2}}{M_2} - \frac{P_{s1} - P_{e1}}{M_1}$$



Multiplied both sides by  $\frac{M_1 M_2}{M_1 + M_2}$

$$\frac{M_1 M_2}{M_1 + M_2} \frac{d^2\delta}{dt^2} = \frac{M_1 M_2}{M_1 + M_2} (P_{s2} - P_{e2}) - [P_{s1} - P_{e1}]$$

$$\frac{M_1 M_2}{M_1 + M_2} \frac{d^2\delta}{dt^2} = \frac{M_1 P_{s2} - M_2 P_{s1}}{M_1 + M_2} - \left( \frac{M_1 P_{e2} - M_2 P_{e1}}{M_1 + M_2} \right)$$

$$P_{aeq} = P_{seq} - P_{eeq}$$

$$P_{aeq} = M_{eq} \frac{d^2\delta}{dt^2}, M_{eq} = \frac{M_1 M_2}{M_1 + M_2}$$

### Swinging of the machines:

$$H_{eq} = H_1 + H_2 + \dots + H_n$$

### Do not swinging of the machines:

$$\frac{1}{H_{eq}} = \frac{1}{H_1} + \frac{1}{H_2} + \dots + \frac{1}{H_n}$$

In general swinging of the machines will prefer so that the equivalent inertia will increase which will reduce the swinging of the machines.

**Eg:**

Generator - 1 having inertia constant  $H_1$  &  $S_2$

Generator - 2 having inertia constant  $H_2$  &  $S_2$

Equivalent inertia constant

$$H_{eq} = \frac{H_1 H_2}{H_1 + H_2}$$

It is not correct, because  $H_1 \neq H_2$ .

Choose a Base i.e., any generator having rating as  $S_{base}$ . To get the swinging

$$H_{1new} = \frac{S_{old} H_{old}}{S_b} \Rightarrow H_{eq} = H_{1eq} + H_{2eq1}$$

$$H_{2new} = \frac{S_{old} H_{old}}{S_b}$$

$$H_{eq} = H_{1New} + H_{2New}$$

For n machine system

$$H_{eq} = \sum_{i=1}^n \frac{H_{iold} S_{iold}}{S_b}$$

### Example 4.2:

A 50 Hz, 4-pole, 500 MVA, 22 kV turbo-generator is delivering rated megavolt-amperes at 0.8 power factor. Suddenly a 3-phase fault occurs at generator terminal, which will be reduce the electrical output by 40% and assume constant power input to the shaft. The accelerating torque of the generator in MNm at the fault will be

**Sol:** Before fault

Mechanical input ( $P_{s1}$ ) = electrical output

$$(P_{e1}) = 500 \times 0.8 \times 10^6 = 400 \text{ MW}$$

During fault electrical output

$$(P_{e2}) = 0.6 \times 400 \times 10^6 = 240 \text{ MW}$$



During fault mechanical input

$$(P_{s_2}) = 400 \text{ MW}$$

Accelerating power,

$$P_a = P_{s_2} - P_{e_2} = 400 - 240 = 160 \text{ MW}$$

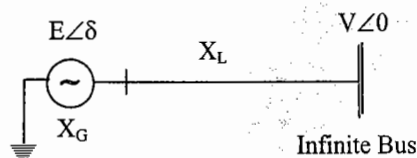
$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$\omega_s = \frac{2\pi N_s}{60} = \frac{2\pi \times 1500}{60} = 157 \text{ rad/sec}$$

Accelerating Torque

$$T_a = \frac{P_a}{\omega_s} = \frac{160 \times 10^6}{157} = 1.018 \text{ M N-m}$$

### Calculation of steady state stability of alternator connected to infinite Bus:



It involves the study of dynamics of rotating body. When the rate of change of load is quite small as compared with the natural frequency of oscillation of the system. The basis is swing equation.

$$M \frac{d^2 \delta}{dt^2} = P_a = P_s - P_e$$

The solution to the above equation can be obtained by linearizing the non-linear differential equation around its initial operating point  $(P_{e_0}, \delta_0)$

$$M \frac{d^2 (\delta_0 + \Delta \delta)}{dt^2} = P_s - (P_{e_0} + \Delta P_e)$$

$$M \frac{d^2 \Delta \delta}{dt^2} = - \frac{\partial P_e}{\partial \delta_0} \Delta \delta$$

$$\left( \frac{M d^2}{dt^2} + \frac{\partial P_e}{\partial \delta} \bigg|_{\delta_0} \right) \Delta \delta = 0$$

$$\frac{d}{dt} = k = \text{Roots of the equation}$$

$$\left( Mk^2 + \frac{\partial P_e}{\partial \delta} \bigg|_{\delta_0} \right) \Delta \delta_0 = 0$$

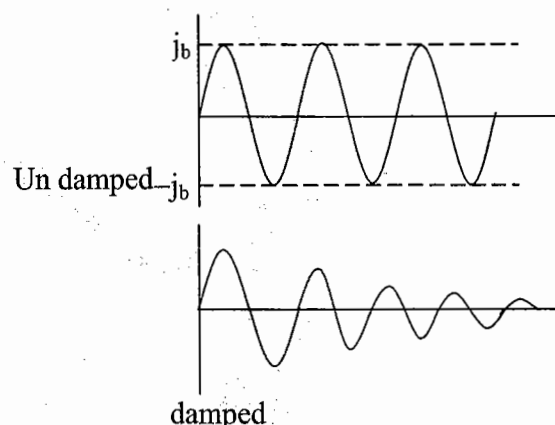
$$K = \pm \left( \frac{1}{M} \left( - \frac{\partial P_e}{\partial \delta} \bigg|_{\delta_0} \right) \right)^{1/2} \text{ rad/sec}$$

$$K = \frac{\text{rad/sec}}{2\pi} = \text{Hz}$$

Case (i):

$$\frac{\partial P_e}{\partial \delta} \bigg|_{\delta_0} = +ve ; \delta_0 < 90^\circ$$

K is pure imaginary conjugate. The roots are undamped oscillations. When compared to angular momentum (M) of the synchronous generator the change of load is less. So the undamped oscillations becomes damped oscillations due to activating of damper windings.



The system is stable

Case (ii):

$$\text{If } \frac{\partial P_e}{\partial \delta} \bigg|_{\delta_0} = -ve ; \delta_0 > 90^\circ$$

K becomes positive. The system unstable

**Example 4.3 :**

A 500 MW, 21 kV, 50 Hz, 3-phase, 2-pole synchronous generator having a rated pf = 0.9, has a moment of inertia of  $27.5 \times 10^3 \text{ kg-m}^2$ . the inertia constant (H) will be

**Sol:** Inertia constant (H)

$$= \frac{\text{Kinetic energy stored in rotor (MJ)}}{\text{MVA rating of alternator (S)}}$$



Kinetic energy stored in rotor =  $\frac{1}{2} I \omega^2$

$$\omega = \frac{2\pi N_s}{60}$$

$$N_s = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

$$\omega = \frac{2\pi \times 3000}{60} = 314.15 \text{ rad/sec}$$

$$\therefore \text{K.E.} = \frac{1}{2} \times 27.5 \times 10^3 \times (314.15)^2$$

$$= 1357 \text{ MJ}$$

$$H = \frac{1357}{\left(\frac{500}{0.9}\right)} = 2.44 \text{ MJ/MVA}$$

### Equal area criteria:

1. It is a graphical solution to the swing equation  
The basic is swing equation.

$$M \frac{d^2 \delta}{dt^2} = P_a = P_s - P_e$$

Multiply both sides the equation  $2 \frac{d\delta}{dt}$  and integrate with respect to time.

$$\int M \frac{d^2 \delta}{dt^2} 2 \frac{d\delta}{dt} dt = \int P_a 2 \frac{d\delta}{dt} dt$$

$$\int \frac{d}{dt} M \left( \frac{d\delta}{dt} \right)^2 = 2 \int P_a d\delta$$

$$M \left( \frac{d\delta}{dt} \right)^2 = 2 \int P_a d\delta$$

$$\frac{d\delta}{dt} = \sqrt{\frac{2}{M} \int P_a d\delta} = 0 \quad (\text{OR}) \quad \int_{\delta_0}^{\delta} P_a d\delta = 0$$

$$\text{at } \frac{d\delta}{dt} \Big|_{\delta_0} = 0 \text{ where } (\omega = \omega_s)$$

$$\text{at } \frac{d\delta}{dt} \Big|_{\delta} = 0 \text{ where } (\omega = \omega_s)$$

2. For transient stability  
For the system to be stable  $A_1 < A_2$   
For the system to be critically stable  $A_1 = A_2$   
For the system to be unstable  $A_1 > A_2$
3. During the swinging of the rotor the maximum swing may exceed  $90^\circ$ . It is only area stability, but not angle stability.

4. For the stability it only consider the first swing of the rotor after disturbance. If it maintain the stability during the first stability it will also maintain the stability during the succeeding if swing. So the chances of losing synchronism is only first swing to calculate

### Assumptions to evaluate the transient stability

- (1) The resistances, the shunt capacitance of generator and transmission lines are ignored. The shunt elements like shunt capacitor (or) shunt inductor at load Bus or generator bus are ignored. The remaining network is represented as transfer reactance.
- (2) The mechanical input is assumed to be constant.
- (3) There is no change in the speed of alternator.
- (4) The damping force provided by damper winding is ignored.
- (5) The voltages behind the reactance's of the Machines are ignored

### Applications of equal area criteria:

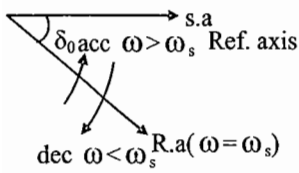
- (i) A sudden increased mechanical input to the synchronous generator.
- (ii) A sudden increased mechanical output on synchronous motor
- (iii) Fault occurs in middle of the line in a parallel transmission s/m
- (iv) Fault occurs on a line near to bus bar on a parallel transmission s/m
- (v) Fault occurs on the bus bar connected to infinite bus of a generator connected to infinite bus by a parallel transmission line.

### Assumptions of equal area criteria:

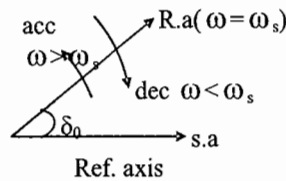
- (i)  $P_a = 0$  and  $\omega = \omega_s$ ; there is no change of rotor angle ie it is called a stable point.
- (ii)  $P_a = 0$  but  $\omega \neq \omega_s$  (or)  $P_a \neq 0$  but  $\omega = \omega_s$  there is a change of rotor angle, it is called swinging point.
- (iii) Consider the moment of inertia of synchronous machine on the swinging.



### Synchronous motor

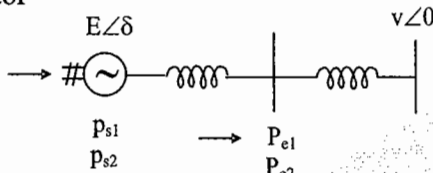


### Synchronous generator



#### Case (i)

Sudden increased Mechanical input to synchronous Generator



$P_{s1}$  = initial mechanical input

$$P_{e1} = \frac{EV}{X_{eq}} \sin \delta_0 \text{ (initial Electrical output)}$$

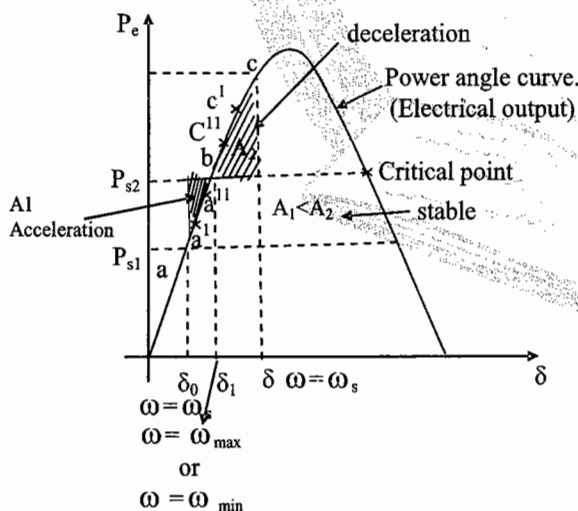
$$P_{e2} = \frac{EV}{X_{2eq}} \sin \delta$$

(after increased Electrical output)

$$X_{eq} = X_g + X_\ell = X_{1eq} = X_{2eq}$$

$P_{s2}$  = increased mechanical input

$$P_{m2} = P_{m1}$$



The synchronous generator is supplied initially with a mechanical input of  $p_{s1}$  and disturbance corresponding electrical output of  $p_{e1}$ , it is represented by point 'a' on power angle curve.

#### Disturbance:-

The mechanical input of synchronous generator will be increased suddenly from  $p_{s1}$  to  $p_{s2}$  in a sudden and large, where as there is no corresponding change in electrical output of synchronous generator because the large inertia of rotating body does not permit a sudden change in rotor angle.

$$\text{Hence } P = P_{s2} - P_{e1}$$

$$= P_{s2} - P_{m2} \sin \delta_0$$

+ve  $\Rightarrow$  acceleration

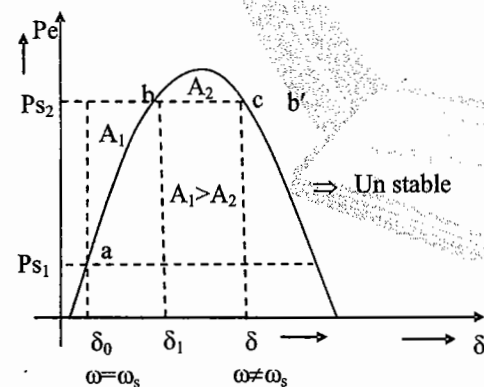
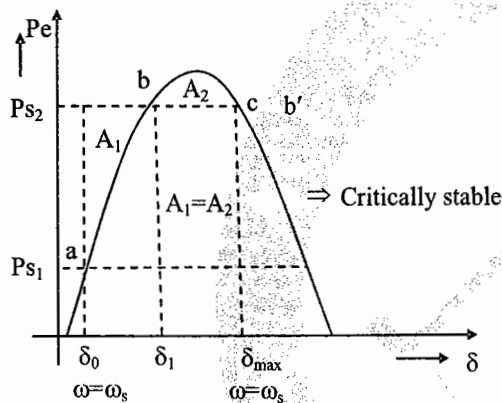
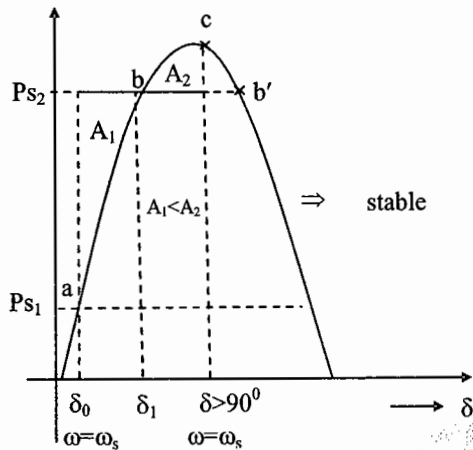
$$\omega > \omega_s$$

The excess mechanical input will be stored in the form of K.E in the rotor in an instantaneous manner.

- The acceleration of synchronous generator will result as the actual speed will be more than synchronous speed which will result as the rotor angle of synchronous generator will be relatively increasing from the previous position. i.e the swinging of generator starts from 'a'
- The angle will increase beyond 'b' in order to get  $\omega = \omega_s$ . If the angle will increase, output is more than mechanical input so that the deceleration will short. The area of deceleration depends on area of acceleration and acceleration depends on mechanical input.
- With the above assumptions it is assumed that point 'c' will be the another swinging point where  $\omega = \omega_s$  is obtained.
- The swinging of m/c beyond 'b' will result as the output will be more than input i.e., stored K.E in rotor will able to convert into electrical output
- Though synchronous machine is swinging, there will be relatively; change in rotor position ie while it is changing from 'a' to 'c', the swinging will start from a new point that is  $a^1$
- Consider the first swing of the m/c for stability analysis.  
It  $\omega = \omega_s$  is obtained before boundary point of  $A_2$ .  $A_1 < A_2$  (stable).
- If the mechanical input is further increased.

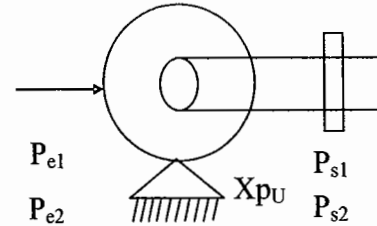


| Position      | Equation                                                                                  | Behavior                                       | Speed                                              | Remarks                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| At 'a'        | $P = P_{s_1} - P_{e_1}$ $= P_{s_1} - P_{m_1} \sin \delta_0$ $= 0$                         | Neither acc<br>nor dec                         | $\omega = \omega_s$                                | there is no change in rotor angle, because it is a single point                                                                       |
| 'a' to 'b'    | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $(\delta > \delta_0)$ $= +ve$   | Acceleration                                   | $\omega > \omega_s$                                | $\delta$ will increase                                                                                                                |
| At 'b'        | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $= 0$                           | Acceleration<br>is zero no<br>deceleration     | $\omega = \omega_{max}$<br>$\neq \omega_s$         | According to (ii)&(iii) assumptions of equal area criteria the rotor angle further increased                                          |
| 'b' to 'c'    | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $\delta > \delta_1$             | deceleration                                   | $\omega < \omega_{max}$<br>but $\omega > \omega_s$ | The machine is having dec but $\omega > \omega_s$ so that rotor angle further increasing                                              |
| At 'c'        | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $= -ve$                         | deceleration                                   | $\omega = \omega_s$                                | It is a swinging point. Further deceleration $\omega < \omega_s$ the angle star decreasing.                                           |
| 'c' to 'b'    | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $\delta < \delta_{max}$ $= -ve$ | Deceleration                                   | $\omega < \omega_s$                                | $\delta$ will reduce                                                                                                                  |
| At 'b'        | $P = P_{s_2} - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta_1$ $= 0$                         | Deceleration<br>is zero and no<br>acceleration | $\omega = \omega_{min}$<br>$\neq \omega_s$         | It is a swinging point due to the effect of moment of inertia, the angle will decrease                                                |
| At 'b' to 'a' | $P = P_2 - P_{e_2}$ $= P_{s_2} - P_{m_2} \sin \delta$ $\delta < \delta_1 (+ve)$           | acceleration                                   | $\omega > \omega_{min}$<br>But $\omega < \omega_s$ | Even though synchronous generator having acceleration, the angle will further decreased because speed is less than synchronous speed. |



- The purpose of equal area criteria is to obtain angle made by rotor at the time of fault cleared by CB which is known as critical clearing angle by considering critically stable condition i.e.  $A_1 = A_2$

## 2. Sudden increased mechanical output on synchronous motor



$$P_{e1} = \frac{EV}{X_{leq}} \sin \delta_0 = p_{m1} \sin \delta_0$$

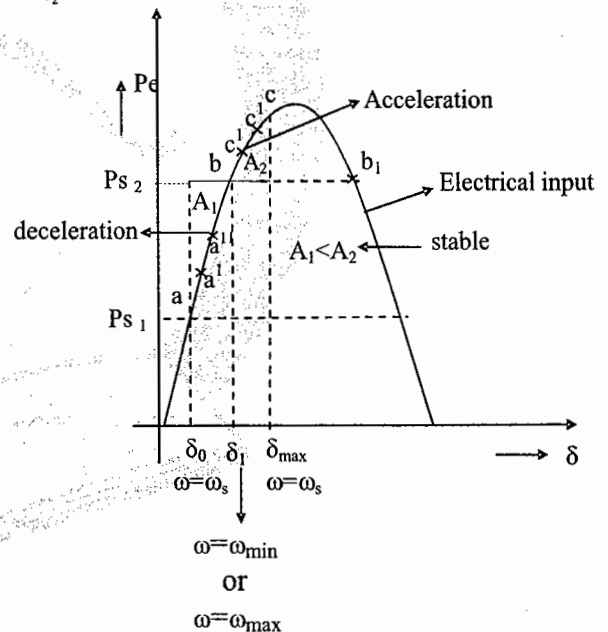
$P_{e2}$  = increased electrical input

$$P_{e2} = \frac{EV}{X_{2eq}} \sin \delta_0 = p_{m1} \sin \delta_0$$

$$p_{m2} = p_{m1}, \quad X_{2eq} = X_{leq}$$

$p_{s1}$  = initial mechanical output

$p_{s2}$  = increased mechanical output



- The equal area criteria synchronous motor is an image of equal area criteria of synchronous generator

$$P = p_e - p_s = -(p_s - p_e)$$

motor = opposite of generator

- The synchronous motor is initially supplied with electrical i/p of  $p_{e1}$  and the corresponding mechanical output of  $p_{s1}$ , it is represented by point 'a' on the power angle curve.



**Disturbance:**

- The load on synchronous motor is increased suddenly from  $p_{s1}$  to  $p_{s2}$  where as there is no corresponding change in the electric input because the inertia of motor does not permit a sudden change in the rotor angle.

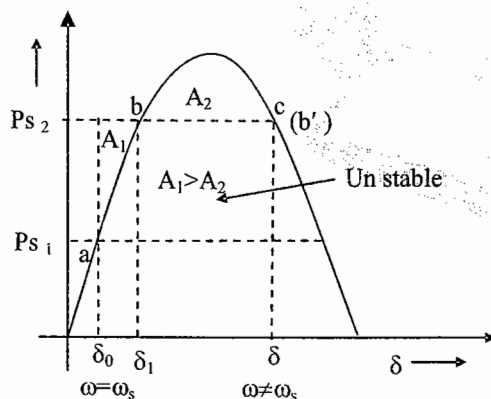
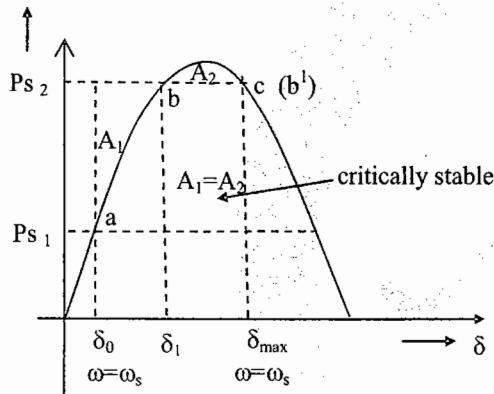
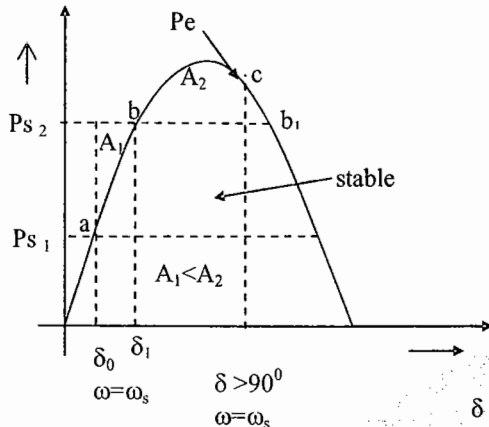
$$\begin{aligned} \text{Hence } p &= p_{e1} - p_{s2} \\ &= p_{m1} \sin \delta_0 - p_{s2} \\ &= -ve = \text{deceleration} \end{aligned}$$

- If the actual speed is not same as syn speed, there will be a relative speed which will result

as the rotor angle will be slowly increases so that the swigging of m/c starts from 'a'

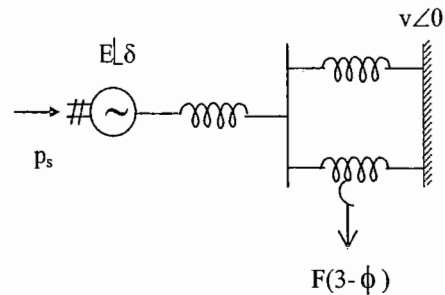
- The angle will increase beyond 'b' in order to get  $\omega = \omega_s$ . As the angle increases.
  - The acceleration will start
  - the area covered beyond 'b' does depends on the area already covered from 'a' to 'b'
- With the above assumptions the swinging of machine is mode up to c to get  $\omega = \omega_s$
- After completing all the oscillations the synchronous machine will be stable at 'b' so that  $\omega = \omega_s$

| Position   | Equation                                                                                             | Behavior                        | Speed                                           | Remarks                                                                                                                         |
|------------|------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| At 'a'     | $p = p_{e1} - p_{s1}$<br>$= \frac{EV}{X_{leq}} \sin \delta_0 - p_{s1} = 0$                           | Neither acc nor dec             | $\omega = \omega_s$                             | there is no change in rotor angle. It is a stable point                                                                         |
| 'a' to 'b' | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta - p_{s2}$<br>$= -ve$                                  | dec                             | $\omega < \omega_s$                             | $\delta$ will increase                                                                                                          |
| At 'b'     | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta_1 - p_{s2}$<br>$= 0$                                  | Dec = 0 and no acceleration     | $\omega = \omega_{min} \neq \omega_s$           | It is a swinging point the angle will increase further due to moment of inertia                                                 |
| 'b' to 'c' | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta - p_{s2} - p_{s2}$<br>$= +ve \quad \delta > \delta_1$ | acc                             | $\omega > \omega_{min}$ but $\omega < \omega_s$ | Even though it is an acc but rotor angle will increase farther because the speed is less than synchronous speed.                |
| At 'c'     | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta - p_{s2}$<br>$= +ve$                                  | acc                             | $\omega = \omega_s$                             | It is a swinging point for further acceleration $\omega > \omega_s$<br>The angle start decreasing the rotor will swinging back. |
| 'c' to 'b' | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta - p_{s2}$<br>$= +ve$                                  | acc                             | $\omega > \omega_s$                             | $\delta$ will decrease                                                                                                          |
| At 'b'     | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta_1 - p_{s2}$<br>$= 0$                                  | Acceleration is zero and no dec | $\omega = \omega_{max} \neq \omega_s$           | Swinging point. The angle will decrease due to effect of moment of inertia.                                                     |
| 'b' to 'a' | $p = p_{e2} - p_{s2}$<br>$= p_{m2} \sin \delta - p_{s2}$<br>$= -ve;$                                 | dec                             | $\omega < \omega_{max}$ but $\omega > \omega_s$ | even though the machine is acc but $\omega > \omega_s$ so that the rotor, angle will further decreased.                         |

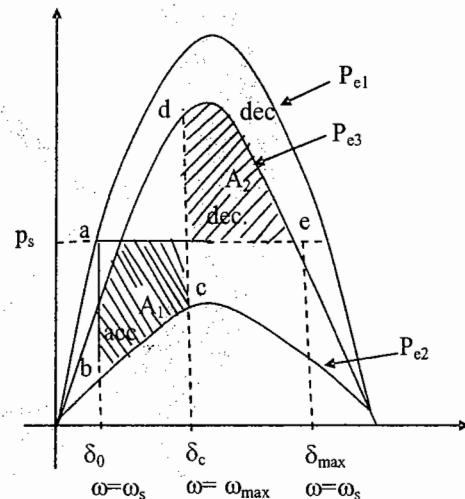


- The purpose of equal area criteria is to evaluate the angle made by the rotor at the line of fault cleared by CB which is called as critical clearing angle by considering critically stable condition i.e.,  $A_1 = A_2$

### 3. Fault occurs at middle of transmission line in a parallel transmission network:-



$p_s$  = mechanical input  
 $p_{e1} = p_{m1} \sin \delta_0 \rightarrow$  before fault  
 $p_{e2} = p_{m2} \sin \delta \rightarrow$  during fault  
 $p_{e3} = p_{m3} \sin \delta \rightarrow$  after fault  
 $p_{m2} < p_{m3} < p_{m1}$   
 $X_{2eq} > X_{3eq} > X_{1eq}$



- The synchronous generator is supplied initially with a mechanical input of  $p_s$  and it is represented by point a on power angle curve.

#### Disturbance:

- A 3- $\phi$  short circuit fault has been taken place in a parallel line at the middle of line suddenly, so that the electrical output is reduced where as the mechanical remain same. It represents by 'b'

$$p = p_s - p_{e2} = p_s - p_{m2} \sin \delta_0$$

$$= +ve = acc (\omega > \omega_s)$$



So that the rotor angle will be starts increasing slowly i.e, the swinging of m/c takes place from 'b'.

- The fault can exist for few cycles, before the fault is cleared by circuit breaker so that the swinging of m/c will be continued till the fault is cleared.
- It is assumed that fault is cleared by breaker at 'c' and the corresponding angle made by rotor is called as critically clearing angle  $\delta_c$

- The faulty line is isolated by breaker so that there will be a sudden change in electric out put due to change in the reactance, the mechanical input is same and it is indicated by point 'd' due to moment of ineletra of rolor the angle will increase future in order to get  $\omega = \omega_s$  in a critically stable manner that is point 'e' on  $p_{e3}$  curve.

### Calculation of $\delta_c$ :

$$\int_{\delta_o}^{\delta_{\max}} p_a d\delta = 0$$

$$\int_{\delta_o}^{\delta_c} p_a d\delta + \int_{\delta_c}^{\delta_{\max}} p_a d\delta = 0$$

$$\int_{\delta_o}^{\delta_c} (p_s - p_{m2} \sin \delta) d\delta + \int_{\delta_c}^{\delta_{\max}} (p_s - p_{m3} \sin \delta) d\delta = 0$$

$$[p_s \delta + p_{m2} \cos \delta]_{\delta_o}^{\delta_c} + [p_s \delta + p_{m3} \cos \delta]_{\delta_c}^{\delta_{\max}} = 0$$

$$[p_s \delta_c - p_s \delta_o + p_{m2} \cos \delta_c - p_{m2} \cos \delta_o] + [p_s \delta_{\max} - p_s \delta_c + p_{m3} \cos \delta_{\max} - p_{m3} \cos \delta_c] = 0$$

$$p_s(\delta_{\max} - \delta_o) + p_{m3} \cos \delta_{\max} - p_{m2} \cos \delta_o = p_{m3} \cos \delta_c - p_{m2} \cos \delta_c$$

$$\therefore \delta_c = \cos^{-1} \left[ \frac{p_s(\delta_{\max} - \delta_o) + p_{m3} \cos \delta_{\max} - p_{m2} \cos \delta_o}{p_{m3} - p_{m2}} \right] \text{elec deg}$$

At 'a'

$$P_s = p_{e1} = p_{m1} \sin \delta_o$$

$$\delta_o = \sin^{-1} \left( \frac{P_s}{p_{m1}} \right) \text{elec deg}$$

$$\delta_{o(\text{read})} = \delta_o \times \frac{3.14}{180}$$

At 'e':-

$$P_s = p_{e3} = p_{m3} \sin \delta_{\max}$$

$$\delta_{\max} = \sin^{-1} \left( \frac{P_s}{p_{m3}} \right) \leq 90^\circ$$

- In a critical stable condition the max swinging angle will be more than  $90^\circ$

$$P_s = p_{e3} = p_{m3} \sin (180 - \delta_{\max})$$

$$\Rightarrow \frac{P_s}{p_{m3}} = \sin(180 - \delta_{\max})$$

$$\Rightarrow 180 - \delta_{\max} = \sin^{-1} \left( \frac{P_s}{p_{m3}} \right)$$

$$\therefore \delta_{\max} = 180 - \sin^{-1} \left( \frac{P_s}{p_{m3}} \right) \text{elec deg}$$

$$\delta_{\max} (\text{rad}) = \delta_{\max} \times \frac{3.14}{180}$$

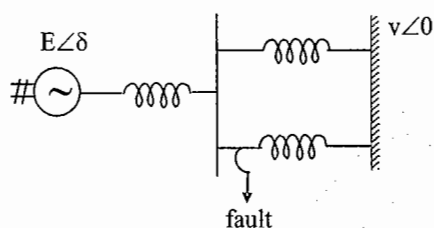
- For any given system there will be a critical clearing angle which is calculated with the help of equal area criteria under the assumption that the s/m is critically stable. However, in order to maintain stability the actual angle made by breaker should be always less than critical clearing angle. Other wise the s/m will loss the stability.

- If the arc will be quenched at a faster rate, the area covered will less before fault is cleared so that the actual angle will be less than critical clearing angle hence the s/m will maintain stability.

$$\text{i.e., } \frac{d\delta}{dt} < 0$$

$$\Rightarrow A_1 < A_2 \text{ stable}$$

#### 4. Fault occurs on a line near to busbar:-



$P_s$  = mechanical input

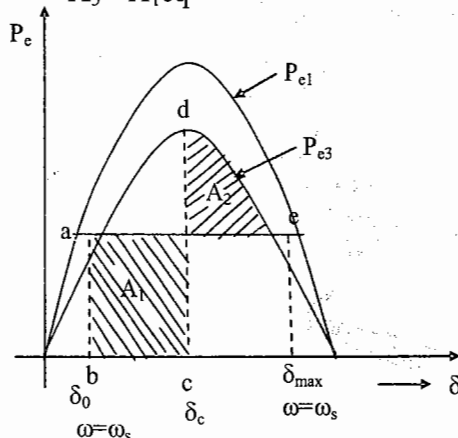
$$P_{e1} = p_{m1} \sin \delta_0$$

$$P_{e2} = 0, p_{m2} = 0$$

$$P_{e3} = p_{m3} \sin \delta$$

$$P_{m3} < p_{m1}$$

$$X_3 > X_{1eq}$$



- The synchronous generator is supplied with mechanical input of  $p_s$  and the corresponding electrical output is  $p_{e1}$  and it is represented at point 'a'
- A 3- $\phi$  fault occurs on one of the parallel line near to busbar and it will become zero where as the mechanical input will remains same and it will be represented by point 'b' on  $\delta$ -axis.

$$\text{Hence } p = p_s - p_{e2} = p_s - 0$$

$$= p_s = +ve \Rightarrow \text{acc}$$

$$\Rightarrow \omega > \omega_s$$

- Due to the acceleration of alt the rotor angle will increase slowly so that the swinging of the m/c will start from 'b' & it will be continued till the fault is cleared by breaker after few cycles and corresponding angle is called critical clearing angle  $\delta_c$ .
- Even though it is assumed as bus fault but the faulty line will isolated by breaker so that the electrical output is increased suddenly due to change in the reactance, where as the mechanical input will remains same and it is represented by point 'd'.
- Due to movement of inertia the rotor angle is further increased even fault is cleared in order to get  $\omega = \omega_s$  in a critical stable manner i.e, point 'e'.

#### Calculation of $\delta_c$ :

$$\int_{\delta_0}^{\delta_c} P_a d\delta + \int_{\delta_c}^{\delta_{\max}} P_a d\delta = 0$$

$$\Rightarrow \int_{\delta_0}^{\delta_c} (p_s - 0) d\delta + \int_{\delta_c}^{\delta_{\max}} (p_s - p_{m3} \sin \delta) d\delta = 0$$

$$\therefore \delta_c = \cos^{-1} \left[ \frac{p_s (\delta_{\max} - \delta_0) + p_{m3} \cos \delta_{\max}}{p_{m3}} \right] \text{elec deg}$$

#### Conversion of $\delta_c$ into $t_c$ :

$$M \frac{d^2 \delta}{dt^2} = p_a = p_s - p_{e2} = p_s \quad [p_{e2} = 0]$$

$$\frac{d^2 \delta}{dt^2} = \frac{p_s}{M}$$

$$\Rightarrow \frac{d\delta}{dt} = \frac{p_s}{M} (t)$$

$$\Rightarrow \delta = \frac{p_s}{M} \frac{t^2}{2} + A \text{ --- constant}$$

$$t \rightarrow t_c \Rightarrow \delta \rightarrow \delta_c$$

$$\Rightarrow \delta_c = \frac{p_s}{M} \frac{t_c^2}{2} + A$$

If  $t_c = 0 \Rightarrow$  the synchronous machine is at initial condition

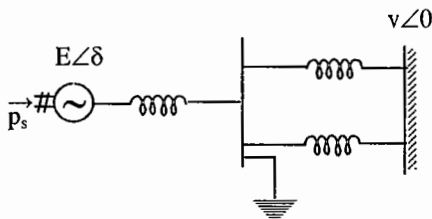
$$\delta_c = \delta_0 = 0 + A \Rightarrow A = \delta_0$$

$$\delta_c = \frac{p_s}{M} \frac{t_c^2}{2} + \delta_0$$

$$\therefore t_c = \sqrt{\frac{2M(\delta_c - \delta_0)}{P_s}}$$

$$t_c \propto \sqrt{M}; t_c \propto \frac{1}{\sqrt{P_s}}$$

### 5. Fault occurs on bus bar:



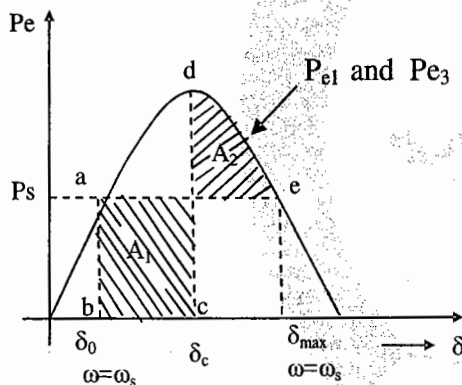
$P_s$  = Mechanical input

$$P_{e1} = P_{m1} \sin \delta_0$$

$$P_{e2} = 0; P_{m2} = 0$$

$$P_{e3} = P_{m3} \sin \delta$$

$$P_{m3} = P_{m1}$$



- If a fault occur on bus bar so that electrical output will become zero, where as the mechanical input will remains same and it will be represented by point 'b' on  $\delta$  axis  
Hence  $p = p_s - p_{e2} = p_s - 0$   
 $= p_s = +ve = acc \Rightarrow \omega > \omega_s$   
i.e, rotor angle start increasing.
- When the fault is cleared by breaker, the alt is isolated from parallel lines, and it will be closed as early as possible, so that the entire system will be restored. When the total system is restored there will be a sudden increase in the output due to change in the reactance of the net work, where as the mechanical input will be remains same, it is represented by point 'd'.

### Calculation of $\delta_c$ :

$$\int_{\delta_0}^{\delta_c} p_a d\delta + \int_{\delta_c}^{\delta_{max}} p_a d\delta = 0$$

$$\int_{\delta_0}^{\delta_c} (p_s - 0) d\delta + \int_{\delta_c}^{\delta_{max}} (p_s - P_{m3} \sin \delta) d\delta = 0$$

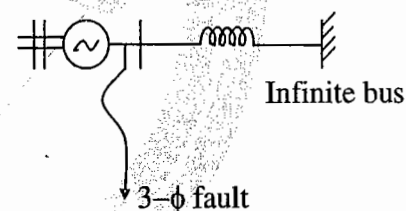
$$\delta_c = \cos^{-1} \left[ \frac{P_s (\delta_{max} - \delta_0) + P_{m3} \cos \delta_{max}}{P_{m3}} \right]$$

$$P_{m3} = P_{m1}$$

### Example 4.4:

A generator delivers power of 1.0 p.u. to an infinite bus through a purely reactive network. The maximum power that could be delivered by the generator is 2.0 p.u. A three-phase fault occurs at the terminals of the generator which reduces the generator output to zero. The fault is cleared after  $t_c$  second. The original network is then restored. The maximum swing of the rotor angle is found to be  $\delta_{max} = 110$  electrical degree. Then the rotor angle in electrical degrees at  $t = t_c$  is

Sol:



$$P_s = P_{e1} = 1.0 \text{ P.u}$$

$$\text{Before fault } P_{max} = 2.0 \text{ P.u}$$

$$\delta_0 = \sin^{-1} \left( \frac{P_s}{P_{m1}} \right) = 30^\circ = 0.523 \text{ rad}$$

$$\text{During fault } P_{e2} = 0 \Rightarrow P_{m2} = 0$$

$$\delta_{max} = 110^\circ = \frac{110 \times \pi}{180} = 1.919 \text{ rad}$$

$$P_{m3} = P_{m1} \quad [\text{Where } P_m = P_{max}]$$

Critical clearing angle,

$$\delta_c = \cos^{-1} \left[ \frac{P_s (\delta_{max} - \delta_0) + P_{m3} \cos \delta_{max} - P_{m2} \cos \delta_0}{P_{m3} - P_{m2}} \right]$$

$$\delta_c = \cos^{-1} \left[ \frac{1.0(1.919 - 0.523) + 2 \cos 110^\circ - 0}{2 - 0} \right]$$



$$\delta_c = 69.14^\circ$$

### Factors Effecting Transient stability:

To maintain the transient stability the area covered during acceleration ( $A_2$ ) in case of generator and vice versa in case of motor.

#### (i) Higher value of 'M'

For a given accelerating power, the higher value of 'M' will result in slower rate of change of rotor angle. So  $A_1 < A_2$ .

#### (ii) Fast acting CB:- (SF6) CB: The fast acting CB will require less cycles to clear the fault so that $A_1 < A_2$ to maintain stability.

#### (iii) High speed Governor:- during the fault condition the real power transfer is reduced, where as the mechanical input is constant. So the alternator will experience acceleration. In order to reduce the acceleration the input also reduced at a faster rate. This can be accomplished by employing high speed governor.

#### (iv) Operating the system at higher operating voltage (or) reducing the net reactance of the system.

This will produce lower initial power angle ( $\delta_0$ ) so that the area for deceleration will be high for the given acceleration.

#### (v) Using fast acting excitation control and fast acting voltage regulator.

During the short circuit fault the voltage at the generator bus also at load bus are reduced which may result in reducing the real power transfer so that the acceleration will increase. To reduce the acceleration the voltage at generator bus and also at load bus are to be improved. This can be possible by having fast acting excitation control at generator and voltage regulator at load bus for unsymmetrical faults.

#### (vi) By using dynamic resistor:

For a line to ground fault on generator the dynamic resistor will increase the real power transfer during the fault so that the acceleration

is reduced. So  $A_1 < A_2$ . The system will maintain transient stability.

Before Resistor  $P_{a1} = P_S - P_{e2}$

After resistor  $P_{a2} = P_S - P_{e2} - I_f^2 R_d$

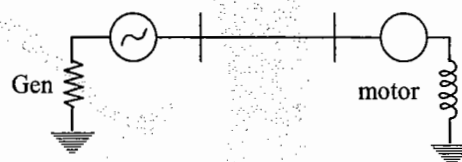
$$P_{a2} < P_{a1}$$

So  $A_1 < A_2$ .

#### (vii) By using single pole CB operation:

Most of the fault at alternator will be single line to ground fault. In case of 3 - phase CB the output power will be zero even for line to ground fault. So the acceleration is high. To limit the acceleration, single pole CB are placed so that the corresponding phase only will trip and the real power transfer will continue through healthy phases.

### Effect of grounding on transient stability



To main transient stability the generator will be ground through resistor because it will increase the real power transfer during the fault. So that the acceleration will reduced.

In case of syn. motor the reactance grounding is preferred which will not affect the real power supplied to the load of motor. So there will not any severe deceleration.